TEXORA · LECTURE SHEET

Introduction to Preparatory Processing

Why grey-state fibre, yarn and fabric cannot be dyed as they are — the natural and added impurities that block coloration, the objectives of preparation, the preparatory sequence each class of textile material follows, and the chemicals that do the work.

TE-2205 · TEXTILE CHEMISTRY & COLORATION-I · TOPIC 1 · LEVEL 2 TERM 2

1. What preparatory processing means

Preparatory processing (also called **pretreatment**) is the set of wet operations that converts grey-state textile material — fibre, yarn or fabric as it leaves spinning, weaving or knitting — into a clean, white, uniformly absorbent substrate ready for dyeing, printing and finishing. Almost every colour defect seen in a finished fabric — unlevel dyeing, resist marks, poor fastness, dull shades — can be traced back to preparation, not to the dyehouse. Preparation is therefore called the foundation of wet processing: dyeing can never be better than the preparation underneath it.

Grey material resists dyeing for two families of reasons:

- **Natural impurities** — substances the fibre itself contains: cotton carries waxes, pectins, proteins, hemicelluloses, ash and seed-coat fragments; wool carries grease (lanolin), suint and dirt; silk carries sericin gum; jute carries lignin and hemicellulose. Several of these are hydrophobic, so they repel the water on which all dyeing depends.
- **Added impurities** — substances put on the material during manufacture: warp **size** (starch, PVA, CMC) applied before weaving, spinning and knitting oils, lubricants, anti-static agents, machine oil stains and handling soil. Size in particular forms a continuous film around the yarn that dye liquor cannot penetrate.

MATERIAL	MAIN IMPURITIES TO REMOVE
Raw cotton (fibre average)	Wax ≈ 0.5–1%, pectin ≈ 0.5–1%, protein ≈ 1–1.5%, ash ≈ 1%, plus motes and seed fragments; total non-cellulosics ≈ 4–8% of fibre weight
Raw wool	Wool grease (lanolin) 5–25%, suint ≈ 5–10%, sand and vegetable matter
Raw silk	Sericin gum ≈ 20–30% coating the fibroin filaments
Woven grey fabric (any fibre)	Warp size (typically 8–15% on cotton warp weight), loom oil, dust
Synthetic fabrics	Spin finishes, knitting oils, oligomers (polyester); no natural impurities

2. Objectives of preparation

- Remove natural and added impurities completely and uniformly, without damaging the fibre.
- Make the material **hydrophilic** — instantly and evenly absorbent to water and dye liquor.
- Achieve a **level, reproducible white** (or clean pastel base) so that dye shades reproduce batch to batch.
- Remove seed-coat fragments, motes and neps that would appear as specks after dyeing.
- Stabilise dimensions and relieve internal stresses (relaxation, heat setting) so later processing is uniform.
- Do all of this with the least water, energy, time and fibre damage the quality target allows.

3. The preparatory sequences

The order of operations is fixed by chemistry: size must come off before scouring liquor can reach the fibre; bleaching only works well on an already scoured, absorbent substrate; mercerisation needs a scoured (often bleached) fabric. The classic cotton woven sequence is:

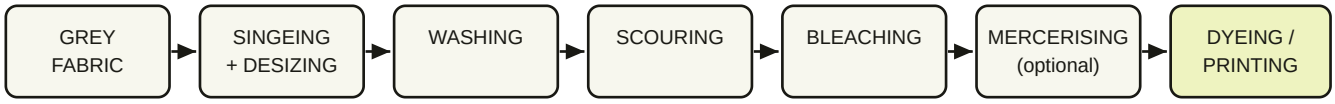


Fig. 1 — Standard preparatory sequence for woven cotton fabric (washing follows each chemical step)

Fig. 1 — The cotton woven preparation line; every chemical stage is followed by washing before the next.

Other materials follow adapted sequences, because their impurities and fibre sensitivities differ:

MATERIAL	TYPICAL PREPARATORY SEQUENCE
Cotton yarn (hank/package)	Scouring → bleaching → (mercerising for high-lustre yarn) → dyeing. No size is present, so no desizing.
Cotton knit fabric	Relaxation → scouring (often combined scour–bleach) → dyeing. Singeing and mercerising are usually omitted.
Wool	Raw-wool scouring (removes grease and suint) → carbonising (acid removal of vegetable matter, when needed) → bleaching (mild, peroxide) → dyeing. Hot strong alkali is avoided — it dissolves wool and causes felting.
Silk	Degumming (sericin removal with soap/enzyme) → bleaching (mild) → dyeing.
Jute	Batching-oil emulsion application and softening → (mild scouring) → bleaching with care → dyeing. Lignin makes jute sensitive to strong bleach and alkali.

Synthetics (e.g. polyester)	Scouring to remove spin finish and oil → heat setting (dimensional stabilisation) → reduction clearing after disperse dyeing where needed. No bleaching step is usually required.
Cellulosic blends (e.g. polyester/cotton)	Desizing → scouring → bleaching (peroxide, safe for both components) → dyeing (two-stage, one per fibre).

4. Chemicals used in preparatory processing

Each step runs on a small cast of chemical families; the detail of each is the subject of Topic 3. The overview:

CHEMICAL FAMILY	EXAMPLES	ROLE IN PREPARATION
Desizing agents	Enzymes (α-amylase), acids, oxidants (persulphate)	Break the size film into soluble fragments that wash away
Alkalis	Caustic soda (NaOH), soda ash (Na_2CO_3)	Scouring: saponify waxes, swell fibre, dissolve pectin and protein
Wetting agents / detergents	Anionic and non-ionic surfactants	Lower surface tension so liquor penetrates rapidly and evenly; lift and suspend soil
Bleaching agents	Hydrogen peroxide, sodium hypochlorite, sodium chlorite	Destroy natural colouring matter (chromophores) to give white
Sequestering (chelating) agents	EDTA, DTPA, phosphonates, polyacrylates	Bind Ca^{2+}, Mg^{2+}, Fe^{3+} so they cannot interfere (e.g. catalyse peroxide damage)
Stabilisers	Sodium silicate, organic stabilisers	Control peroxide decomposition during bleaching
Acids	Acetic acid, sulphuric acid (carbonising)	Neutralisation after alkali steps; pH adjustment
Enzymes	Amylase, pectinase, cellulase, catalase	Mild, specific, low-temperature preparation (desizing, bio-scouring, biopolishing, peroxide clean-up)

5. Glossary — key terms

- **Grey (greige) goods:** fabric as it comes from the loom or knitting machine, unfinished.
- **Size:** protective film (starch or synthetic polymer) applied to warp yarn before weaving.

- **Preparation / pretreatment:** all operations between grey state and coloration that clean, whiten and stabilise the material.
- **Absorbency:** the speed and uniformity with which the material takes up water; the single most practical quality check of preparation.
- **Liquor ratio (material : liquor):** the weight of bath per weight of goods, e.g. 1 : 10.

6. Key points

- Preparation removes natural impurities (wax, pectin, protein, grease, sericin) and added impurities (size, oils) so the substrate is clean, white and absorbent.
- Sequence order is chemically fixed: desize → scour → bleach → (mercerise) → dye.
- Each fibre adapts the sequence to its impurities and its sensitivity to alkali, heat and oxidants.
- Most downstream colour faults originate in preparation; absorbency is its master quality variable.

7. Summary

Preparatory processing is the bridge between grey goods and coloration. Grey material is loaded with natural impurities from the fibre and added impurities from spinning and weaving, and these block wetting and dye uptake. A planned sequence — singeing, desizing, scouring, bleaching and, where specified, mercerising, with washing between stages — strips them away using alkalis, surfactants, oxidising bleaches, enzymes and sequestering agents, each material class following the variant of the sequence its chemistry allows. The result aimed at is always the same: a uniformly clean, white, absorbent and damage-free substrate.

8. Practice

Answer in your own words; brief model answers follow.

1. A mill dyes grey cotton fabric directly after desizing, skipping scouring. Predict two dyeing faults and explain their cause.
2. Why is desizing placed before scouring in the woven-cotton sequence, and why is there no desizing step for yarn?
3. Wool is never scoured in hot, strong caustic soda. Give two chemical reasons.
4. List the preparatory sequence for (a) polyester fabric and (b) silk fabric, naming what each step removes.
5. A polyester/cotton woven fabric is to be prepared. State the sequence and one constraint the blend imposes on the bleaching agent.

Answer key

1. Unlevel/patchy dyeing and poor rubbing fastness with dull shades: cotton wax and pectin remain, so the fabric is still hydrophobic — dye liquor wets and penetrates unevenly and dye sits on the surface instead of bonding uniformly.
2. The size film coats the warp yarn and shields the fibre from scouring liquor; until it is removed, NaOH cannot reach the waxes and pectins. Yarn was never sized (size is a weaving aid applied to warp only), so yarn preparation starts at scouring.
3. (1) Wool keratin is a protein with alkali-labile disulphide and peptide bonds — hot strong NaOH hydrolyses and dissolves it. (2) Heat, alkali and agitation together cause felting (scale-driven irreversible shrinkage), destroying the fabric.
4. (a) Polyester: scouring to remove spin finish/knitting oil → heat setting to stabilise dimensions → dyeing; no natural impurities and no bleaching need. (b) Silk: degumming to remove sericin (soap or enzyme) → mild peroxide bleaching → dyeing.
5. Desize → scour → peroxide bleach → dye. Constraint: the bleach must not damage either component — hydrogen peroxide is used; hypochlorite regimes safe for cotton alone may tender the blend and chlorite needs care, so peroxide is the standard blend-safe choice.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib